

SMART INDIA HACKATHON 2026



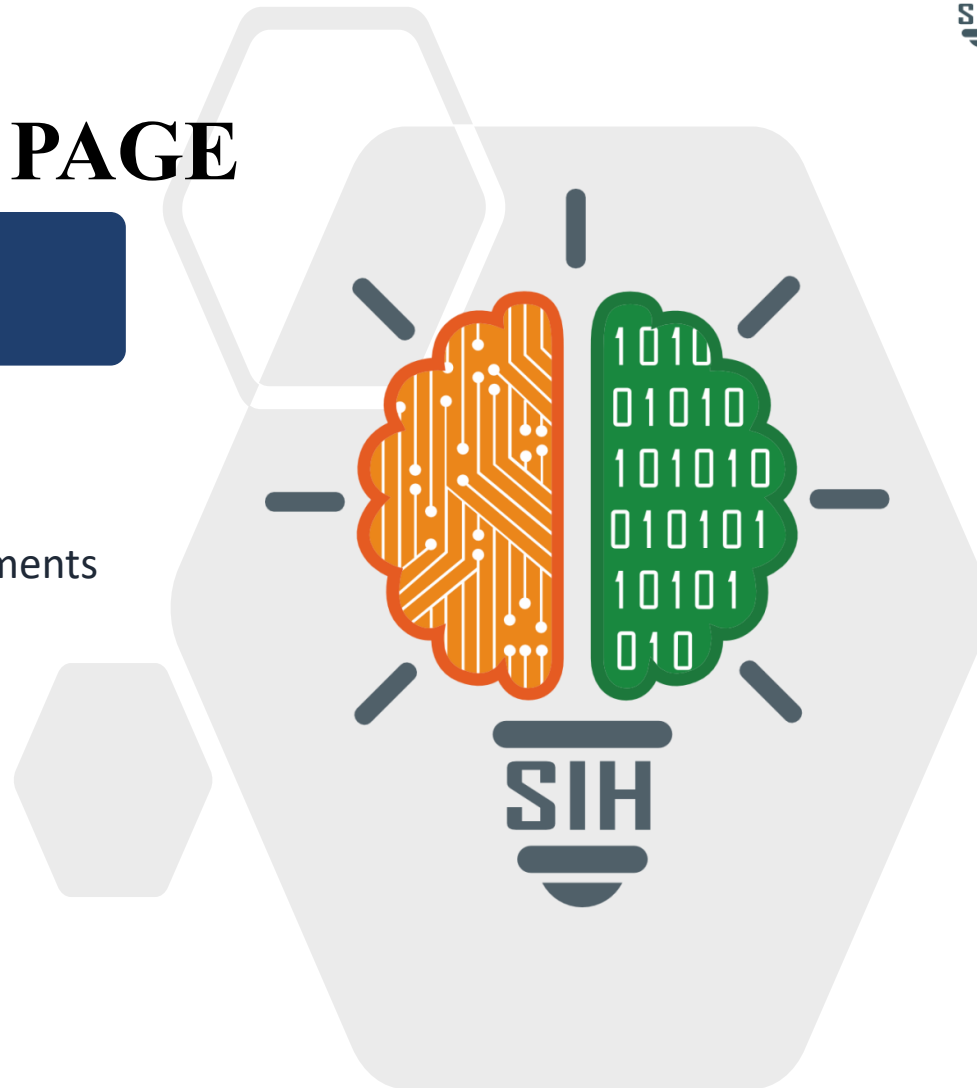
TITLE PAGE

TULA Unified Digital Lifecycle Platform for Legal Metrology

- **Problem Statement ID** – 26036
- **Problem Statement Title** – Development of an Online Verification System for Weighing and Measuring Instruments
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID** – 182718
- **Team Name** – FriendlyFire

One instrument → one digital ID → one certificate anyone can check

Live prototype: tula-legal-metrology.vercel.app





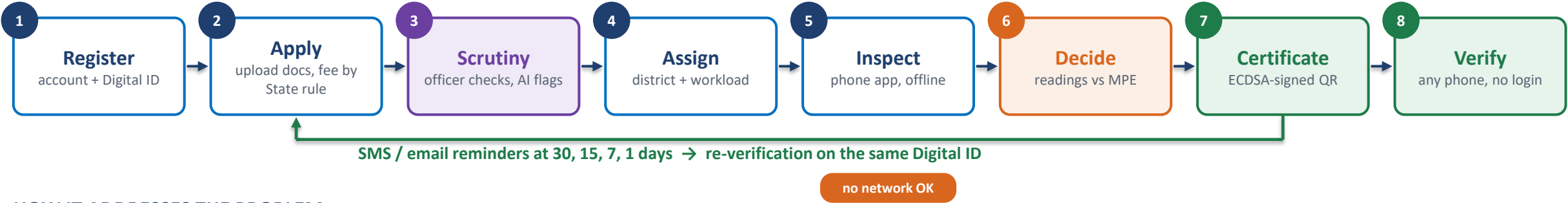
TULA: ONE INSTRUMENT, ONE DIGITAL ID



PROPOSED SOLUTION

A live web + mobile platform: every instrument gets a permanent Digital ID, and application, scrutiny, field inspection (works offline), signed QR certificate, reminders and enforcement all attach to it.

DETAILED EXPLANATION: the life of one instrument (built and running)



HOW IT ADDRESSES THE PROBLEM

Paper records in isolated systems
→ **One live database for every State**

Manual scheduling, pendency
→ **Enforced workflow + pendency dashboard**

Fake or copied certificates
→ **Signed QR: edited copies fail the scan**

Missed re-verification dates
→ **Automatic SMS / email reminders**

INNOVATION AND UNIQUENESS

Unfakeable QR
Server-signed (ECDSA P-256). Checked on any phone, even offline, plus a live revocation list

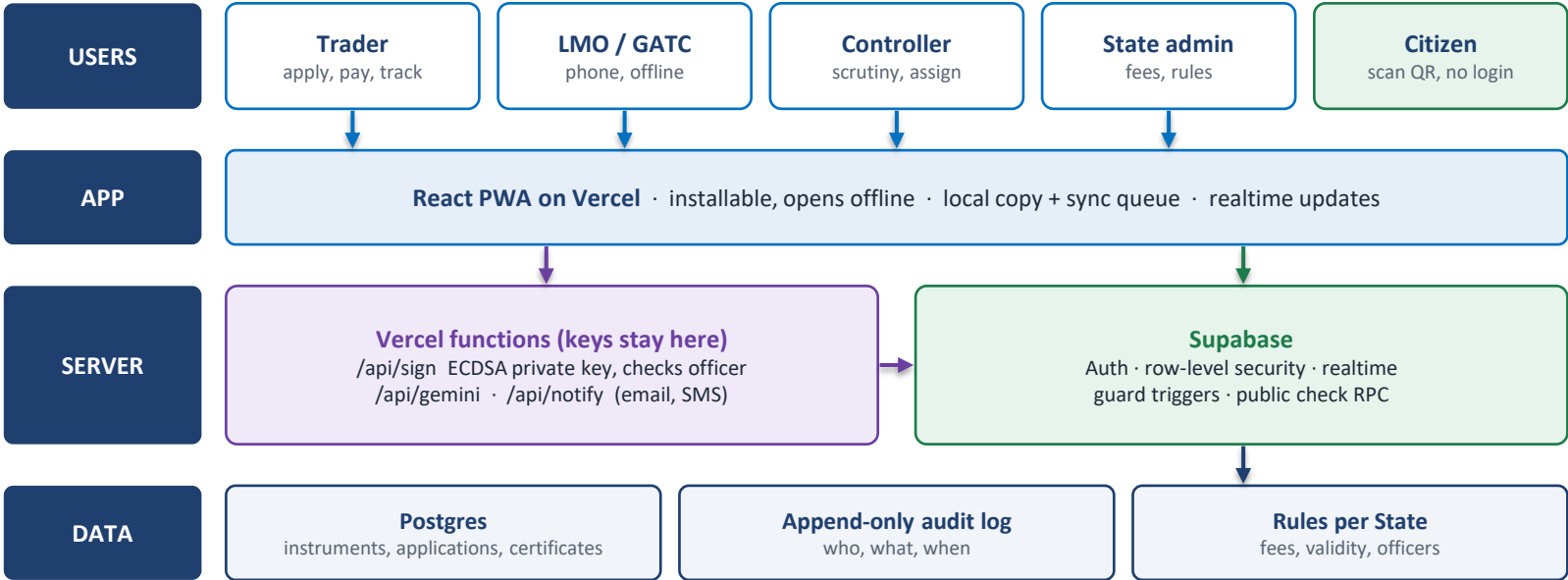
Offline field app
Installable app. Inspection saves on the phone and syncs, signs and notifies on reconnect

Computed verdict
Readings are checked against OIML MPE limits. PASS is impossible unless every check passes

Multi-State by design
Fees, validity and officers are data per State. Database rules isolate each State

AI copilot, human decides
Gemini flags odd documents. It never approves, prices or passes anything

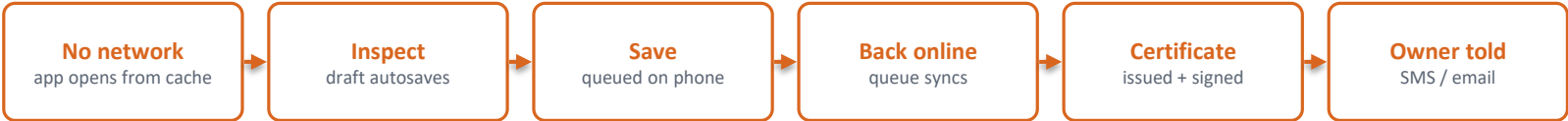
METHODOLOGY AND PROCESS FOR IMPLEMENTATION (as deployed)



HOW A FAKE QR IS CAUGHT



OFFLINE FIELD INSPECTION



TECHNOLOGIES USED

- React 19 · TypeScript · Vite · Tailwind installable, mobile-first
- Service worker · local store · sync queue offline work, auto sync
- Supabase Postgres · RLS · Auth · Realtime live, per State
- Vercel functions · ECDSA P-256 · jsQR sign on server, check on phone
- Google Gemini (Flash) document flags, help chat
- Resend · Twilio · jsPDF email, SMS, certificate PDF

ON A PHONE, FROM THE LIVE PROTOTYPE

Genuine

Edited copy

Expired

Revoked

"Edited copy" takes the genuine certificate and changes only its expiry date. The signature no longer matches, so the check fails.

Fake or edited certificate

The signature does not match the certificate details. It was edited or forged. It claims to be DL/LM/2026/08912. Report it so the district office can inspect.

Weight or reading looks wrong, seal broken, or certificate expired?

Report a problem

Buyer scans an edited copy: rejected

Zero load

PASS

Test load

0.000 kg

Error

0.000 kg

Reading (kg)

0.000

Limit (MPE)

± 0.003 kg

25 % of Max

FAIL

Test load

7.500 kg

Error

+0.012 kg

Reading (kg)

7.512

Limit (MPE)

± 0.005 kg

50 % of Max

PASS

Test load

15.000 kg

Error

0.000 kg

Reading (kg)

15.000

Limit (MPE)

± 0.008 kg

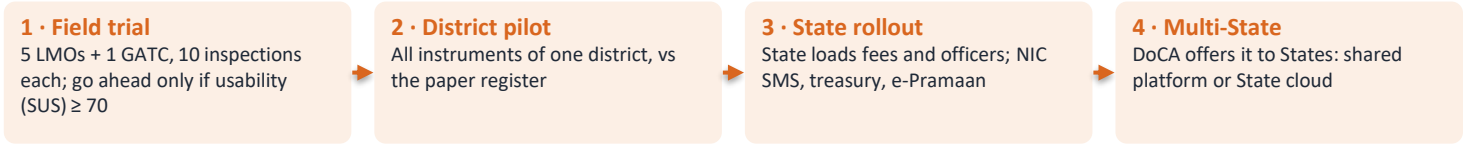
Over the legal limit: flagged at once



ANALYSIS OF THE FEASIBILITY OF THE IDEA: what is built vs planned

PS requirement	Live today	Next
Registration, profiles, secure login	✓ real accounts, 6 roles	e-Praamaan, phone OTP
Apply for (re-)verification	✓ own instruments, docs, fee	treasury payment
Scheduling, allocation LMO / GATC	✓ district + workload	officer calendars
Inspection results recorded	✓ MPE computed, photos, GPS	cloud photo storage
QR certificate + authentication	✓ signed QR, revocation	HSM-held key
Validity tracking, alerts	✓ 30/15/7/1-day reminders	nightly server job
SMS / email updates	🕒 email + SMS wired, keys pending	NIC SMS gateway
Dashboards: pending, inspections	✓ stage, ageing, district	state-wide wallboard
Search, export, print	✓ search, PDF, CSV	bulk audit export
Mobile app for field officers	✓ installable, offline	Android package

HOW WE TEST WITH REAL OFFICERS, THEN SCALE



VIABILITY: HOW IT GROWS, SURVIVES AND IS FUNDED

Grow in the market
Buyers are DoCA and the 36 State / UT Legal Metrology departments; traders and citizens use it free. Each State that adopts is the reference for the next.

Survive
Open-source, pay-per-use cloud, no licence fees. Running cost fits in the service charge States already add to verification fees. No lock-in: standard Postgres, can move to State cloud.

Get funded
SIH award and incubation first, then Startup India Seed Fund or MeitY Startup Hub grants to reach the pilot; State e-governance budgets pay for rollout as a work order.

POTENTIAL CHALLENGES AND RISKS → STRATEGIES

No network at site	Offline app; syncs, signs and notifies later
Fake / copied QR	Signed QR + revocation list; serial shown on scan
Rules differ by State	Fees and validity are data per State
Data privacy (DPDP 2023)	Row-level security, minimal public fields
AI mistakes	Advisory only; officer decides; logged
Officer adoption	Field trial first; plain home screen per role; text size, contrast, Hindi



Complete live demo

Scan for the guided demo: register → apply → officer inspects (try offline) → signed certificate → scan the QR on another phone. Each step ticks itself.

Evaluator password: TulaDemo@2026

tula-legal-metrology.vercel.app/demo



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

LMOs & GATCs

- Jobs on the phone, even offline
- Pass / fail calculated for them
- Certificate issued on site

Businesses

- Apply and pay without a visit
- SMS / email at every step
- Reminder before a stamp expires

Consumers

- Scan any certificate QR
- Fakes and edits are rejected
- Report a problem in two taps

Government

- Pending by stage, age, district
- Inspections due and overdue
- Full audit trail per instrument

MEASURED ON THE LIVE SYSTEM (30 Sep 2026, scripted end-to-end run on the production database)

2 min 51 s

application filed → signed certificate (system time, 3 people, no paper)

6 s

certificate issued → signed by the server

6 / 6

forged or edited QR copies rejected (automated tests)

0

paper forms and office visits in the run

PILOT TARGETS vs TODAY'S PAPER BASELINE (baseline measured from the pilot district's register)

≤ 7 days

application → certificate (median)

< 3 s

certificate check by QR scan

100%

due instruments reminded 30 days early

0

paper certificates in the pilot district

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

Social

Fair weight at shops, pumps and mandis. Buyers can check.

Economic

Fewer visits and faster renewals for traders and MSMEs.

Environmental

Paperless forms, certificates and records.

Governance

One auditable trail from application to enforcement.

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK



References & Links

- LM Act 2009; LM (General) Rules 2011 · consumeraffairs.gov.in
- LM (GATC) Rules 2013; State Enforcement Rules
- OIML R 76-1 (scales), R 117-1 (dispensers) · oiml.org
- IT Act 2000, DPDP Act 2023
- Supabase, Vercel and Gemini API docs

COMPARED WITH TODAY

	Paper	Typical State portal	TULA
Certificate	copyable paper	editable PDF	signed QR
Buyer can check	no	rarely	any phone, offline
Field record	register	typed later	phone, offline
Pass / fail	judgement	typed	computed vs MPE
Across States	separate	one per State	one platform



Scan: a real certificate
 DL/LM/2026/08913, issued and signed on the live system on 30 Sep 2026. Your phone checks the signature.
 Code: github.com/prerak2507/tula-legal-metrology